

ADAPTER DESIGN PATTERN

- ① Definition
- ② Where to use Adapter Design Pattern
- ③ Understand Adapter
- ④ Understand Adapter Design Pattern
- ⑤ Practical for Adapter Design Pattern
- ⑥ Use Factory Design Pattern with Adapter Design Pattern

Defⁿ

- ① Structural design pattern
- ② It acts bridge between two incompatible interfaces, allowing them to work together.

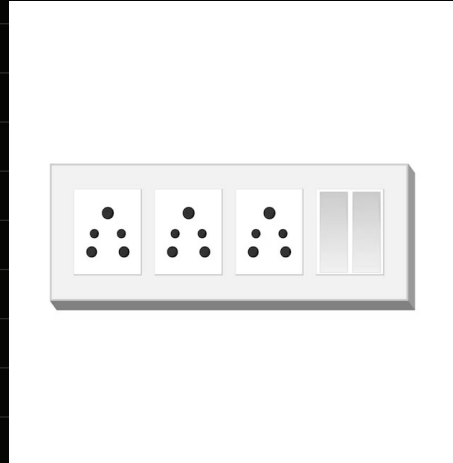
Where to use

- ① If there is a requirement where we need to interact with 3rd party API/SDK, then we can think for adapter design pattern.

What is Adapter?



US Charger



Indian Socket

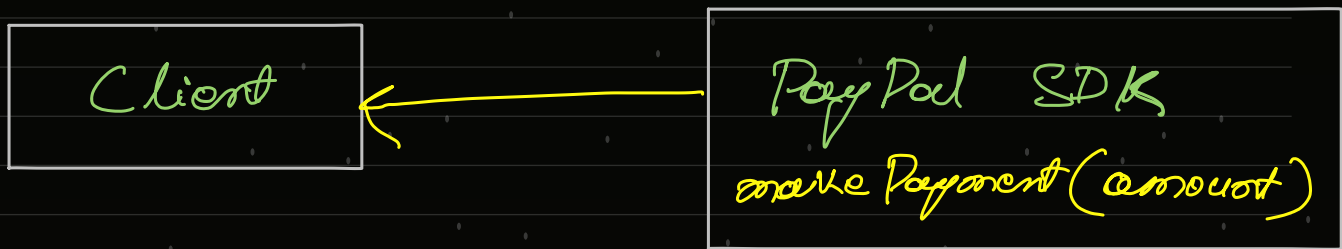


Adapter

Understand Adapter Design Pattern

Requirement-1 [Release-1]

- ① In your application, you need integrate PayPal payment gateway.



```
class Client {  
    void pay(int amount) {  
  
        PayPalSDK palSDK = new PayPalSDK();  
        palSDK.makePayment(amount);  
  
    }  
}
```

Requirement-2 (Release-2)

- ① You need^{to} support Razonpay payment gateway.

Client

Razorpay SDK
payment(amount)

```
class Client {  
    void pay(int amount) {  
  
        PayPalSDK palSDK = new PayPalSDK();  
        palSDK.makePayment(amount);  
  
    }  
}
```

~~PayPalSDK~~ ~~palSDK~~ = new Razorpay SDK;
palSDK.payment(amount);

Problem with these solutions

① Violating

→ SRP

→ OCP

→ DIP

SOLID

② Tightly coupled code.

How to bind?

